

TM Forum Introductory Guide

Agent to Agent Protocol for Telecoms

IG1453

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Executive Summary

The TM Forum AN functional architecture standard IG1251C defines a three-layer hierarchical structure. Within this architecture, each layer and domain rely on agents to perform AN tasks and achieve closed-loop management. Additionally, interactions between agents across different layers, domains, and services are necessary. To address multi-agent interaction challenges, the industry has proposed the A2A protocol. Considering the key features and specific requirements of the ICT field, this document introduces an extended Agent-to-Agent for Telecoms protocol. This protocol is tailored for AN scenarios, facilitating effective multi-agent communication and collaboration. This extension aims to enhance interoperability and efficiency in complex telecommunications environments by providing a standardized interaction framework among agents across various layers and domains.

1. Introduction

This guide defines a multi-agent interaction framework for the ICT domain, designed to meet the requirements of cross-layer, cross-domain, and cross-service multi-agent interaction for operation and management tasks in autonomous networks. Specifically, it includes:

1. The definition of a task description data model named Task-T for the ICT domain, which isolates differences in task processing among built-in LLMs of different agents, improves task interaction efficiency, and balances the capabilities of natural language processing in abstracting tasks with the accuracy demands of task execution in the ICT field.
2. The definition of a network event subscription and reporting mechanism, allowing network-layer agents to proactively report network status updates, thereby supporting the dynamic correction capabilities required for network task processing. This mechanism can also serve as an active trigger for cross-service scenario tasks.
3. The definition of a multi-level negotiation mechanism for tasks, providing negotiation protocols for task ambiguity, feasibility, and dynamic information correction, along with resource conflict management capabilities required for ICT domain tasks.

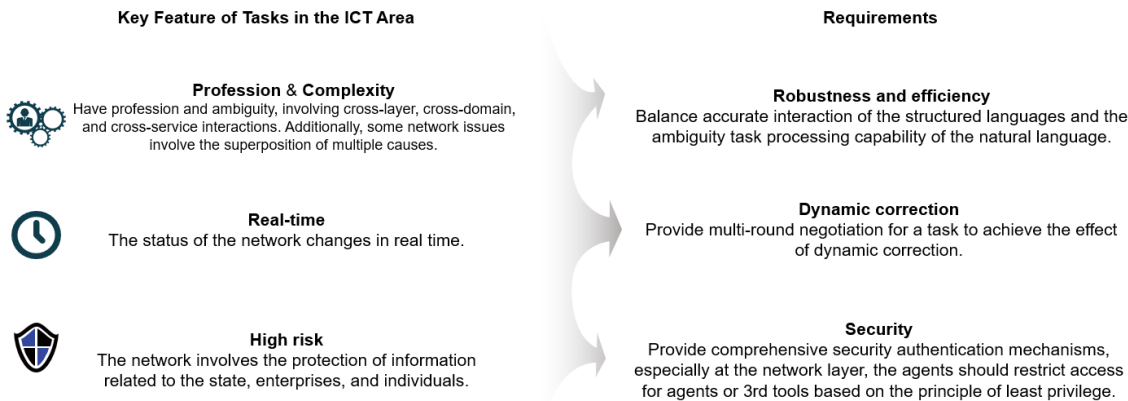
Note: This guide does not modify the A2A protocol defined by the Linux Foundation. It defines payloads and interaction models that define agent communication content for telecom operation tasks. This definition can be carried over A2A or other agent interaction protocols. This document uses A2A as the implementation to present extension examples and reference code only (Further extensions are FFS).

1.1. Key feature and requirements for A2A in ICT Area

In the ICT field, tasks exhibit several common characteristics and requirements.

1. **Profession & Complexity:** ICT tasks are highly specialized and complex. For example, in fault management, wireless faults alone can be categorized into 15 major types with 189 root causes. At the same time, not all faults generate alarms, which requires specialized diagnosis and repair capabilities. The interaction among functional modules must be accurate and efficient. Additionally, network services involve extensive cross-layer, cross-domain, and cross-service interactions, requiring robust solutions for handling complex and ambiguous tasks.
2. **Real-time:** these tasks demand real-time processing. Phenomena such as network traffic tidal effects, holidays, major events, and hotspot incidents cause rapid changes in network status, necessitating real-time adaptive corrections to enhance response efficiency.

3. **Security:** Networks deal with vast amounts of privacy and sensitive data. As agent capabilities grow, ensuring their security and trustworthiness becomes essential. Especially, the use of agentic AI solutions has introduced numerous new attack vectors, and efforts are still underway to find solutions that ensure security within the operational paradigm. This requires not only basic security authentication but also mechanisms to log agent-involved tasks and collaborative activities, enabling traceability and accountability.



2. Scenarios

The AN architecture follows a design logic based on a three-layer hierarchical structure with single-domain task closed-loop control and cross-domain task collaboration. Its typical agent interactions include:

- 1) Cross-layer interactions, such as those between the Business layer, Service layer, and Network layer, exemplified by complaint ticket processing.
- 2) Cross-domain interactions (across technical domains or management and maintenance domains, etc.), for example, a typical cross technical domain scenario is fault management involving interactions between agents from the wireless technology domain and agents from the backhaul network technology domain.
- 3) Cross-service interactions, such as when a fault occurs, trigger service assurance to restore the affected services and adjust energy-saving strategies to ensure a consistent user experience., involving agent collaboration across different services. These specific scenarios are thoroughly described in IG1251E.
- 4) Human-Agent Interaction: Used to coordinate and avoid decision-making conflicts, assessing whether human involvement is necessary for coordination or if it can proceed without human intervention.

3. Gap Analysis

As outlined above, in the ICT domain, agent communication protocols must meet the following requirements:

1. Ensuring the security and traceability of agent interactions
2. Providing task-based multi-round negotiations to enable dynamic corrections and address real-time network issues
3. Balancing the accuracy of structured language with the efficiency of natural language processing to handle ambiguous tasks.

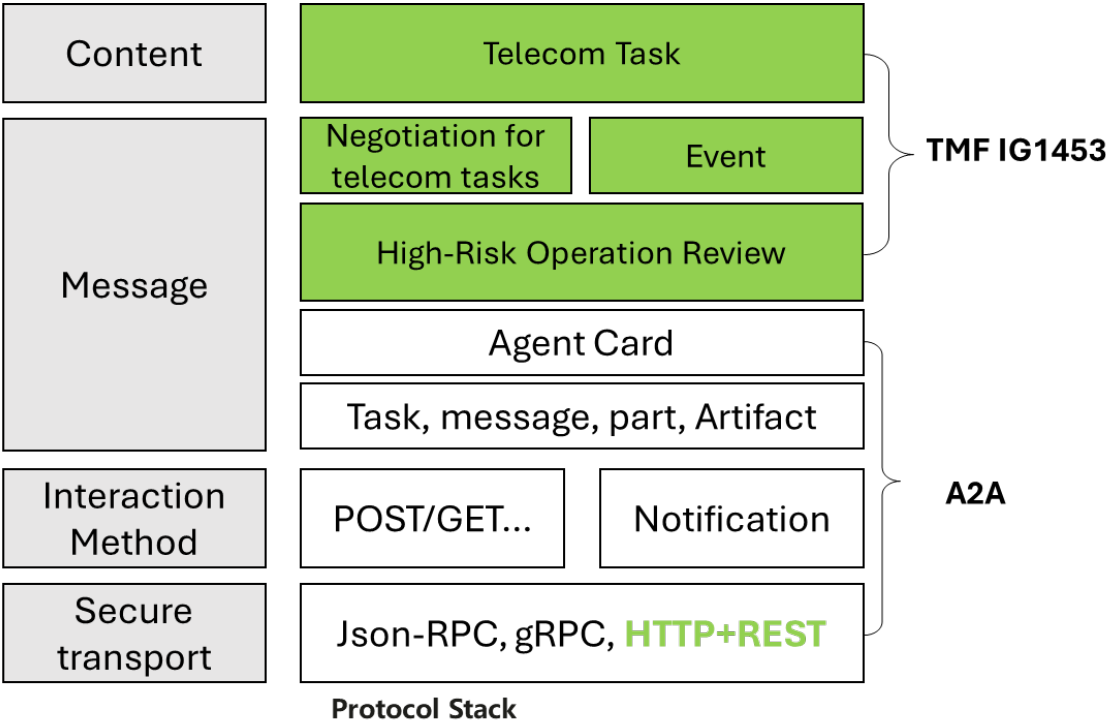
These criteria are essential to effectively address the professional and complex challenges inherent in ICT tasks.

This document analyzes the gap between the capabilities of the A2A protocol and the interaction requirements of agents in the ICT domain, as shown in the table below.

Requirements	A2A
Registration & Discovery	√
Agent Description	√
Task Status	√
Task Processing	√
Efficiency and professional prompt and self-explanatory terminologies in ICT area	•JSON format, supporting text, structured data, and multimodal data.
Task negotiation to provide dynamic correction capabilities	•Only support negotiation of user interface capabilities, such as iframes, videos, and web forms
Transmission Protocol, TMF support HTTP+RESTFUL	√
Subscription & Notification	√
Security	•supports various authentication methods including HTTP Basic Auth, API Key, and OAuth 2.0.

4. Protocol Stack

Based on the above analysis, we believe that the Agent to Agent protocol for Telecoms should adhere to HTTP+RESTful standards at the transport layer. The Interaction Method layer can follow the existing basic operations of A2A. At the message layer, in addition to inheriting the current functionalities and capabilities of A2A such as Task, Message, Part, Artifact, and Agent Card, it is necessary to extend it to include secure and traceable mechanisms, as well as task-based multi-round negotiation capabilities. Furthermore, the content layer should define a set of structured prompt templates to balance the accuracy of structured language with the efficiency needed for natural language processing in handling ambiguous tasks.



5. Extension methods

A2A supports four extension mechanisms: 1) Data-only, 2) Profile, 3) Method (Extended Skills), 4) State Machine. Extensions only take effect for the agent that declares them and do not support universal extensions that can be used by all agents. The use of extensions requires activation declaration in the Header to be enabled (<https://a2a-protocol.org/latest/topics/extensions/>). A typical A2A extension method follows these steps:

Step 1: Extension Declaration

An agent declares its support for an extension by including an AgentExtension object within the AgentCapabilities object of its Agent Card.

```
{
  "name": "Magic 8-ball",
  "url": "https://example.com/agents/eightball",
  "capabilities": {
    "streaming": true,
    "extensions": [
      {
        "uri": "https://github.com/a2aproject/a2a-samples/samples/extensions/timestamp/v1",
        "description": "Adds a creation timestamp to returned Message and Artifact objects in the 'metadata' field.",
        "required": false
      }
    ]
  }
}
```

Step 2: Extension Specification Documentation

The specific behavior and structure of an extension are defined by its specification. While there is no mandatory fixed format, it includes the following:

- A unique URI that identifies the extension.
- The schema and meaning of the object within the AgentExtension.params field.
- The schema for any additional data structures used in communication between the client and the agent.
- New request-response flows.
- Other logical details required to implement the extension.

Step 3: Extension Activation & Usage

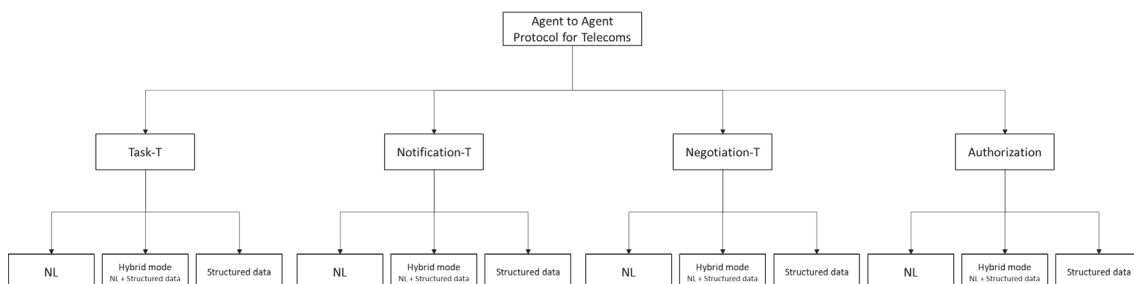
- Request: The client requests to activate extensions by including the X-A2A-Extensions header in the HTTP request.
- Response: Once the agent has confirmed all activated extensions, the response MUST contain the X-A2A-Extensions header listing all extensions that were successfully activated for that request.

```
POST /agents/my-agent HTTP/1.1
Host: example.com
Content-Type: application/json
X-A2A-Extensions: https://github.com/a2aproject/a2a-samples/samples/extensions/timestamp/v1
Content-Length: ...
```

6. Protocol description

The protocol for network tasks can be defined as encompassing task allocation, synchronous/asynchronous task processing, negotiation of task objectives, dynamic adjustments during task execution, and negotiation of task outcomes. These aspects are specifically described in IG1251E. This paper primarily focuses on the extensions of the protocol. For the content inherited from the A2A protocol, please refer to A2A (<https://a2a-protocol.org/latest/specification/>). The LF A2A protocol operates at the protocol layer, defining the structure and exchange of agent messages independent of any interface or API definition. This specification extends A2A at the same layer, adding telecom-specific structure for task description, event notification, and negotiation. TM Forum Open APIs operate at the interface layer and are not in scope for this specification

To achieve AN L4, the industry has defined a series of standardized interfaces (including intent), such as the TM Forum's intent interface TMF921 and 3GPP's intent interface as defined in 3GPP TS 28.312. To facilitate AN L4 evolution and rapid deployment, the protocol shall support any of the following forms: structured natural language prompts, structured data defined by telecom SDOs (including 3GPP, TM Form, and IETF, etc), or a hybrid of both. This fully aligned with A2A which supports various inputs, including natural language and other multimodal inputs.



This document defines **specific extensions** to a generic A2A interaction model for telecom operations. This draft specifies four interoperable profiles:

- **Telecom Task Extension (Task-T)** — tasking and execution for telecom intents.
- **Event Subscription and notification (Notification-T)** — event subscription, filtering, and delivery.
- **Task Negotiation Extension (Negotiation-T)** — multi-turn proposal for task terms.
- **Task Authorization Extension (Authorization-T)** — Review/Approval of high-risk operations and dynamic permission application.

Agent to Agent protocol for Telecoms should share one **high-level, technology-neutral information model** and can bind to NL, 3GPP, or IETF while preserving meaning in a common envelope.

6.1. Telecom Task

Unlike general multi-agent interactions that may serve a wide range of activities in the natural world, multi-agent interactions in autonomous network management have several distinct characteristics. Firstly, these scenarios are relatively well-defined and focused, dedicated solely to network management and operation tasks. For example, the TMF has identified twenty high-value scenarios specifically prioritized for AN Level 4 development. Under conditions where the scenarios and tasks are clearly defined, structured language enables more effective and accurate interaction between agents. Secondly, interactions among agents in autonomous network management often involve ambiguous or fuzzy tasks. This includes coordinating across multiple domains and dealing with multitask collaboration, as well as handling user-initiated complaints that are typically expressed in natural language. In such cases, network agents need to understand user needs expressed in natural language and translate them into actionable tasks. Therefore, multi-agent interactions in autonomous networks require an “interaction language” capable of balancing the ambiguity and expressiveness of natural language with the precision of structured languages. This enables agents to communicate efficiently and accurately, ensuring effective execution of complex, often fuzzy, tasks.

Structured data offer accuracy and stability advantages for deterministic task execution and precise information specification. The telecom task model defined in this paper supports multiple representation forms, including NL-based structured templates and structured data formats.

This specification focuses on the domain information model, instantiating Task content through the telecommunications-domain structured prompt to meet the flexibility requirements of telecommunications tasks.

6.2. Compatibility with Intent data model

In theory, A2A can support various inputs, including natural language and other multimodal inputs. Embedding published standard interfaces within the content section is technically doable.

To achieve AN L4, the industry has defined a series of standardized interfaces, such as TM Forum's Intent Interface (TMF921) and intent interfaces defined by 3GPP TS 28.312. These standards have been deployed, making backward compatibility an unavoidable issue. A2A also offers extension methods, such as including standardized intent interfaces within message part. However, the practical effectiveness and whether efficient parsing can be achieved remain to be

validated. This guide focuses on natural language-based Network task interaction, and compatibility with intent interfaces.

Agent to Agent protocol for Telecoms inherits A2A's extension mechanism by carrying telecom data models in the Message. Natural-language task descriptions and data models can be placed in the metadata block, while released structured data models, such as 3GPP 28.312, can be carried in the data block. Through above extension mechanism, this protocol effectively supports widely adopted structured data models and enables hybrid use of natural language and structured data.

6.3. Telecom Task Extension

6.3.1. Overview

This capability has been extended based on the A2A protocol, aiming to enhance the accuracy of task processing while maintaining flexibility.

The expansion of tasks in the telecommunications domain is carried out based on the principle of maximizing the reuse of existing A2A mechanisms. An overview of the expanded interfaces for tasks in the telecommunications field is outlined below.

A Task-T message MAY be expressed in natural language or reference 3GPP specification.

When using natural language (NL), reference the following URI.

URI:

" <https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1> "

When adopting the 3GPP approach, align with TS 28.312 and use the following URI:

URI:

" https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/3GPP_Intent/v19.0 ", which includes the URIs for 3GPP TS 28.312 R19 specification and intent YAML schema as follows:

URI for 3GPP TS 28.312 :

<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3554>

URI for 3GPP R19 intent YAML schema: " https://forge.3gpp.org/rep/sa5/MnS/-/blob/Rel-19/OpenAPI/TS28312_IntentNrm.yaml?ref_type=heads "

According to the A2A extension method, the Agent Card should be extended first. The relevant definition example in Task-T-NL Extension is as:

```
"name": "fault-mgmt-agent",
"description": "Fault Management Agent",
"url": "http://0.0.0.0:8085",
"version": "1.0",
"defaultInputModes": [
  "text"
],
"defaultOutputModes": [
  "text"
],
"capabilities": {
  "streaming": "true",
  "push_notifications": "true",
  "extensions": [
    {
      "uri": "https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1"
      "uri": "https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/3GPP_Intent/v19.0"
      "description": "Add the TASK-T extension to the 'metadata' field of the sent Message object."
    }
  ]
},
"skills": [
  {
    "id": "fault",
    "name": "Fault Diagnosis",
    "description": "Fault detection and root cause diagnosis"
  }
],
"supportsAuthenticatedExtendedCard": true
}
```

6.3.2. Data Model

6.3.2.1. Natural language data model

In AN L4, TMF has currently identified and summarized 20 high-value scenarios, which are expected to evolve gradually into cross-scenario and cross-service comprehensive network intelligent scenarios in the future. Accordingly, the task model for AN L4 should aim to handle high-value scenario tasks. The data model needs to be based on these high-value scenarios, abstracting common foundational structures, rules, and relationships shared among various scenario models to enhance the accuracy of agents in handling specialized tasks.

Additionally, leveraging the flexibility of natural language can improve the robustness in processing abstract tasks.

This document analyzes the fault management, complaint handling, performance optimization, and energy efficiency optimization tasks within the currently defined high-value scenarios of AN L4. The metamodel for network tasks (Task-T) is defined as following.

- **Task-T data model attributes**

- **Task Description** is used in messages sent from the sender agent (the task requester) to the receiver agent (the task executor) to describe the fundamental requirements of the task. Its values are expressed in natural language or structured data. In practical applications, this parameter can directly contain specific or abstract task information. When the task details and requirements are sufficiently concrete, the receiver agent can understand/parse the task intent along with the necessary associated information and related resources for execution, and the task will be carried out. When the task information and requirements are relatively abstract, the sender agent needs to provide relevant details in parameters such as Task Type, Task Object, and Environment Information to facilitate the receiver agent the parsing and execution of the task.
- **Task Type** is an optional attribute used to define the type of the task. For task types already defined in the current AN L4, such as fault management, complaint handling, performance optimization, etc., they can be directly filled into this parameter. For unknown scenarios, such as cross-scenario interactions or other abstract tasks, this attribute can be omitted. Its value is described in natural language or structured data.
- **Task Object** is an optional attribute used to define the goal of the task and related information, such as the incident received in a fault management scenario or the complaint ticket number in a complaint handling scenario. This attribute is not limited to the above examples. For unknown scenarios, such as cross-scenario interactions or other abstract tasks. Its value is described in natural language or structured data.
- **Task Context** is an optional attribute used to define the environmental context related to the task, such as the time and location of a fault occurrence in a fault management scenario, or additional contextual information like neighboring management domains or related agent details. This attribute is not limited to the above examples. For unknown scenarios, such as cross-scenario interactions or other abstract tasks, its value can be omitted or described from other perspectives. Its value is described in natural language or structured data.

- **Constraints** is an optional attribute used to define the priority of the task to help resolve conflicts among task objectives. It is recommended to use descriptions such as high, medium, low, etc., but no strict restrictions are imposed. Its value is described in natural language or structured data.
- **Expected Output** defines the format of the anticipated result. It is recommended to include whether the task was successful, records of task operations, and suggested actions for manual intervention, but no strict restrictions are imposed. Its value is described in natural language or structured data.

Task Prompt Template for Telecom Extension Specification

The structure and style of prompts play a crucial role in guiding AI agents to perform tasks effectively. In AN L4, for well-known high-value scenario tasks, using highly reliable prompt templates can significantly enhance task execution efficiency. In response, Task-T prompt templates are defined based on the Task-T data model for different scenarios. This section introduces four examples of prompt templates for fault management, energy efficiency optimization, complaint handling, and performance optimization. Future versions will feature optimizations and provide additional efficient prompt templates. Considering the high readability, structured organization, and ease of format conversion of Markdown, which make it widely used in the field of generative AI, this article utilizes Markdown format to introduce the metamodel sample. The specific format is as follows: using "##" as the identifier for metamodel properties, followed by entering the name of the metamodel property. All metamodel extension examples in this article are expressed in Markdown format, and no further elaboration will be provided in subsequent sections.

An example of a **Fault Repair Task Prompt Template** is shown below and should be documented in the specification document following the :

Task Description

Please resolve the fault based on the following fault information.

Task Type

Fault Repair

Task Object

Includes elements such as a list of fault serial numbers.

Task Context

Includes elements such as the time of occurrence, location, and fault context.

Constraints

Includes elements such as priority and the content of the task result output.

Expected Output

The fault resolution result should include the following information:

1. Resolution Result: Success or Failure
2. Fault Resolution Details
3. Reason for Modification Failure (if applicable)
4. Domain-Specific Information

An example of a **Complaint Handling Task Prompt Template** is shown below and should be documented in the specification document:

Task Description

Please analyze the root cause of the complaint and provide restoration suggestions based on the following complaint information.

Task Type

Complaint Handling

Task Object

Includes elements such as the complaint work order number.

Task Context

Includes elements such as the time of occurrence and complaint context.

Constraints

Includes elements such as task priority.

Expected Output

Defines the result return of the task response. The result return includes:

1. Diagnostic Status: Success or Failure
2. Diagnostic Conclusion
3. Restoration Suggestions
4. Root Cause List of the Complaint
5. Domain-Specific Information

An example of an **Energy Efficiency Optimization Task Prompt Template** is shown below and should be documented in the specification document:

Task Description

Please analyze the root cause of the complaint and provide restoration suggestions based on the following complaint information.

Task Type

Energy Efficiency Optimization

Task Description

Please output an energy consumption optimization solution and provide the optimization effectiveness based on the following energy-saving targets and energy efficiency information.

Task Object

Includes the unique intent ID, optimization object definition (object type, instance, context), and optimization targets (energy consumption metrics, throughput, etc.)

Task Context

None

Constraints

Includes elements such as priority.

Expected Output

Defines the result return of the task response.

1. Overall optimization status (Success, Partial Success, or Failure)
2. Optimization result for each target, including the report ID, intent reference ID, intent satisfaction information, and expected execution result list (optimization actions and status)
3. Domain-specific information

An example of a **Performance Optimization** Task Prompt Template is shown below and should be documented in the specification document:

Task Description

Please analyze the root cause of the complaint and provide restoration suggestions based on the following complaint information.

Task Type

Energy Efficiency Optimization

Task Description

Please output an energy consumption optimization solution and provide the optimization effectiveness based on the following energy-saving targets and energy efficiency information.

Task Object

Includes the unique intent ID, optimization object definition (object type, instance, context), and optimization targets (energy consumption metrics, throughput, etc.)

Task Context

None

Constraints

Includes elements such as priority.

Expected Output

Defines the result return of the task response.

1. Overall optimization status (Success, Partial Success, or Failure)
 - a. Optimization result for each target, including the report ID, intent reference ID, intent satisfaction information, and expected execution result list (optimization actions and status)
2. Domain-specific information

6.3.2.2. 3GPP Intent Mode

3GPP TS 28.312 defined a generic intent (represents one or multiple IntentExpectation(s) which includes MnS Consumer's requirements, goals and contexts) and scenario specific IntentExpectation(s), for examples:

- RadioNetworkExpectation, which can be used to represent MnS Consumer's expectations for radio network delivering and performance assurance.
- RadioServiceExpectation, which can be used to represent MnS Consumer's expectations for radio service delivering and assurance in the specified area.

The JSON/YAML schemas (including TS28312_IntentNrm.yaml, TS28312_IntentExpectations.yaml) for 3GPP intent s can be found in the https://forge.3gpp.org/rep/sa5/MnS/-/tree/Rel-19/OpenAPI?ref_type=heads.

6.3.3. Usage Example for Telecom Task Extension

6.3.3.1. Overview

This chapter provides three use cases, each describing the definition method and instantiated data for natural language tasks, structured data tasks, and hybrid-mode tasks. The aim is to help readers understand how to use the above three task modes, enabling them to design and develop data models and corresponding templates for tasks related to specific scenarios according to their own needs.

6.3.3.2. Usage Example for fault repair with natural language

This chapter introduces how to activate and utilize the telecommunications domain task extension.

An additional header field, X-A2A-Extensions, needs to be included in the request header, with the value: https://

projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1

One example of request body are shown as following:

```

https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1
Body:
{
  "message": {
    "role": "agent",
    "parts": [
      {
        "text": "Fix the base station service outage in City X on January 16."
      }
    ],
    "metadata": {
      "https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1": "## Task Description\nPlease repair the fault based on the following fault information.\n## Task Type\nFault Repair\n## Task Object\nFault serial number list: 150285, 488703\n## Task Context\nLocation: X City\nTime: 2024-01-16 16:56:32\nFault Context: 4G base station outage fault.\n## Constraints\n\n## Expected Output\nThe fault resolution result should include the following information:\n1. Resolution Result: Success or Failure\n2. Fault Resolution Details\n3. Reason for Resolution Failure\n4. Domain-Specific Information\n",
      "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
    }
  }
}

```

The response header contains X-A2A-Extensions:

[https:// projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1](https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1)

One example of response body are shown as following:

```

{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts":

```

```

{
  "artifactId": "artifact-uuid",
  "parts": [
    {
      "text": "The fault diagnosis task has been completed.",
    }
  ]
  "metadata":
  {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1": "Diagnosis status: Succeeded.\n2. Fault diagnosis and analysis result description: NE: P33206-YWHJ-** Service aggregation equipment room 1 (990) port: 12-TPJ1EM8F-4(**)2024/10/24) received no optical signal because the peer device was powered off or the port was closed. \n3. Suggestion: The fault may be caused by an internal fault of the NE. Contact NE technical support engineers to locate the fault. \n4. Root cause list:\n Root cause 1: \n Fault name: Peer equipment is powered off.\n Fault object ID: 3600HWCSCEPRT1c2c318e9c54e3d2a\n Root cause details of the fault object: NE: P33206-YWHJ-** Port (990) in service aggregation equipment room 1:12-TPJ1EM8F-4(**)2024/10/24) does not receive optical signals because the peer device is powered off or the port is closed. \n Repair suggestion corresponding to the root cause: The fault may be caused by the NE internal fault. Contact the NE technical support engineers to locate the fault. \n Fault object type: PRT\n Fault root cause location: P33206-YWHJ-** Service aggregation equipment room 1 (990) port: 12-TPJ1EM8F-4\n n5. Domain-specific information: None.\n"}
  }
}

```

6.3.3.3. Usage Example for RAN energy saving with structured data

This chapter introduces how to utilize the telecommunications domain task extension with structured data for RAN energy saving.

An additional header field, X-A2A-Extensions, needs to be included in the request header, with the value:

["https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/3GPP_intent/v19.0"](https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/3GPP_intent/v19.0)

One example of request body are shown as following:

```
{
  "messageId": "9229e770-767c-417b-a0b0-f0741243c589",
  "message": {
    "role": "user",
    "parts": [
      {
        "data": {
          "Intent": {
            "Id": "Intent_5",
            "userLabel": "RAN_Energy_Saving",
            "intentExpectation": [
              {
                "expectationId": "1",
                "expectationVerb": "Ensure",
                "expectationObjects": [
                  {
                    "objectInstance": "SubNetwork_1",
                    "objectContexts": [
                      {
                        "contextAttribute": "CoverageAreaPolygon",
                        "contextCondition": "IS_ALL_OF",
                        "contextValueRange": [
                          {
                            "convexGeoPolygon": [
                              { "latitude": "31.2696", "longitude": "121.6322" },
                              { "latitude": "31.2668", "longitude": "121.6323" },
                              { "latitude": "31.2669", "longitude": "121.6412" },
                              { "latitude": "31.2696", "longitude": "121.6410" }
                            ]
                          }
                        ]
                      }
                    ]
                  }
                ]
              }
            ],
            "expectationTargets": [
              {
                "targetName": "RANEnergyConsumption",
                "targetCondition": "IS_LESS_THAN",
                "targetValueRange": "1000"
              },
              {
                "targetName": "AveULRANUETHpt",
                "targetCondition": "IS_GREATER_THAN",
                "targetValueRange": "100"
              }
            ],
            "expectationContexts": [
              {
                "contextAttribute": "TargetAssuranceTime",
```



```

        "fulfilmentStatus": "FULFILLED",
        "targetAchievedValue": "100"
      },
    ]
  }
]
},
"intentConflictReports": [
  {
    "conflictType": "TARGET_CONFLICT",
    "conflictingTarget": "AveDLRANUETHpt",
    "recommendedSolutions": "MODIFY"
  }
]
}
}
}
}
]
}
]
}
}

```

6.3.3.4. Usage Example for radio service delivery with a hybrid of NL and structured data

This chapter introduces how to utilize the telecommunications domain task extension with hybrid of NL and structured data for radio service delivery.

An additional header field, X-A2A-Extensions, needs to be included in the request header, with the value: " <https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/NL/v1>", "https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/3GPP_intent/v19.0"

One example of request body is shown as following:

```

{
  "messageId": "9229e770-767c-417b-a0b0-f0741243c589",
  "message": {
    "role": "user",
    "parts": [
      {
        "text": "A large-scale music event will be held at Jing'an Temple Square from 19:30 to 22:30 on November 26, 2025. During the performance period, 20 fixed and mobile video capture devices are deployed to support live streaming, which requires the

```

mobile network to provide high requirements on both latency and bandwidth for uplink transmission under mobility conditions."

```
    },
    {
      "data": {
        "Intent": {
          "Id": "intent_8",
          "userLabel": "Radio_Service_Deliver",
          "intentExpectation": [
            {
              "expectationId": "1",
              "expectationVerb": "Deliver",
              "expectationObjects": [
                {
                  "objectType": "Radio_Service",
                  "objectContexts": [
                    {
                      "contextAttribute": "ServiceType",
                      "contextCondition": "IS_EQUAL_TO",
                      "contextValueRange": ["eMBB"]
                    },
                    {
                      "contextAttribute": "UEGroup",
                      "contextCondition": "IS_ALL_OF",
                      "contextValueRange": [
                        {"pLMNId": "46000"},
                        {"sNSSAI": "1"}
                      ]
                    }
                  ]
                }
              ]
            }
          ],
          "expectationTargets": [
            {
              "targetName": "ULLatency",
```

```
    "targetCondition": "IS_LESS_THAN",
    "targetValueRange": "15"
  },
  {
    "targetName": "DLLatency",
    "targetCondition": "IS_LESS_THAN",
    "targetValueRange": "20"
  },
  {
    "targetName": "ULThptPerUE",
    "targetCondition": "IS_GREATER_THAN",
    "targetValueRange": "100"
  },
  {
    "targetName": "DLThptPerUE",
    "targetCondition": "IS_GREATER_THAN",
    "targetValueRange": "300"
  },
  {
    "targetName": "numberOfUEs",
    "targetCondition": "IS_LESS_THAN",
    "targetValueRange": "500"
  }
]
}
]
}
}
]
```

One example of response body is shown as following:

```
{
  "task": {
    "id": "f7c8d9e0-1234-4abc-8d12-abcdef987654",
    "status": {
      "state": "completed"
    },
    "artifacts": [
      {
        "artifactId": "a1b2c3d4-5678-4e9f-8abc-123456abcdef",
        "parts": [
          {
            "text": "The eMBB radio service delivery task associated with Intent_8 has been successfully executed. All service delivery targets were successfully met within the specified schedule and coverage area."
          },
          {
            "data": {
              "IntentReport": {
                "id": "Radio_Service_Delivery_Report",
                "intentReference": "Radio_Service_Deliver",
                "lastUpdatedTime": "2025-11-13T19:45:00Z",
                "intentFulfilmentReport": {
                  "intentFulfilmentInfo": {
                    "fulfilmentStatus": "SUCCESS"
                  },
                  "expectationFulfilmentResults": [
                    {
                      "expectationId": "1",
                      "targetFulfilmentResults": [
                        {
                          "targetName": "ULLatency",
                          "targetFulfilmentInfo": {
                            "fulfilmentStatus": "FULFILLED"
                          },
                          "targetAchievedValue": "12"
                        },
                        {
                          "targetName": "DLLatency",
                          "targetFulfilmentInfo": {
                            "fulfilmentStatus": "FULFILLED"
                          },
                          "targetAchievedValue": "18"
                        },
                        {
                          "targetName": "ULThptPerUE",
                          "targetFulfilmentInfo": {
                            "fulfilmentStatus": "FULFILLED"
                          },
                          "targetAchievedValue": "110"
                        }
                      ]
                    }
                  ]
                }
              }
            }
          }
        ]
      }
    ]
  }
}
```

```

    {
      "targetName": "DLThptPerUE",
      "targetFulfilmentInfo": {
        "fulfilmentStatus": "FULFILLED"
      },
      "targetAchievedValue": "320"
    },
    {
      "targetName": "numberOfUEs",
      "targetFulfilmentInfo": {
        "fulfilmentStatus": "FULFILLED"
      },
      "targetAchievedValue": "480"
    }
  ]
}
]
}
}
}
}
]
}
]
}
}

```

6.4. Network Event Subscription and Notification

6.4.1. Overview

The AN L4 provides two modes for automatic network task closed-loop management: Proactive mode and Reactive mode. Proactive mode refers to a bottom-up approach where the network automatically detects issues, resolves problems, or triggers corresponding tasks. Reactive mode, on the other hand, is a top-down process initiated by engineers or customers, where the Business Layer Agent, Service Layer Agent, and Network Layer Agent execute tasks in a closed loop.

For the Proactive mode, upper-layer agents such as the Business Layer Agent or Service Layer Agent need to sense events occurring at lower layers, like the Network Layer Agent, for example, the occurrence of faults. This enables the triggering of related tasks, such as cross-domain fault management or energy-saving tasks. Additionally, during network task execution, since the network is in real-time flux, events related to the tasks also need to be perceived by upper-layer agents.

To address above issues, this protocol has established a structured prompt-based network event subscription and reporting mechanism. This mechanism ensures real-time perception of network events and provides consistent task descriptions across different layers, domains, through structured prompts. This approach balances technical specificity with robustness in handling abstract tasks.

The brief overview of this capabilities extension is as follows:

- **Capability Extension:** telecom-event-subscription-notification-metadata
- A Notification-T message **MAY** be expressed in natural language or reference 3GPP specification:
 - a. When using natural language (NL), reference the URI: projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/NL/v1
 - b. When adopting the 3GPP approach, use the URI: https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/3GPP_Intent/v19.0

Agent Card JSON Code

```
{
  "name": "fault-mgmt-agent",
  "description": "Fault Management Agent",
  "url": "http://0.0.0.0:8085",
  "version": "1.0",
  "defaultInputModes": [
    "text"
  ],
  "defaultOutputModes": [
    "text"
  ],
  "capabilities": {
    "streaming": "true",
    "push_notifications": "true",
    "extensions": [
      {
        "uri": "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/NL/v1"
        "uri": "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/3GPP_Intent/v19.0"
      }
    ]
  }
}
```

```
"description": "Add the Notification-T extension to the 'metadata' field of the Message  
object being sent."  
}  
],  
,  
"skills": [  
{  
  "id": "fault",  
  "name": "Fault Diagnosis",  
  "description": "Fault detection and root cause diagnosis"  
}  
],  
"supportsAuthenticatedExtendedCard": true  
}
```

6.4.2. Notification-T Data Model

At AN L4, the proactive handling of tasks by the Network Layer Agent and the reporting of network events are fundamental capabilities that are widely employed. In AN interaction scenarios, network event reporting can serve as a trigger for related tasks or be used to dynamically gather and update relevant information during task execution by the Service Layer Agent.

This section analyzes the incident events, performance optimization events, and energy efficiency optimization events within the high-value scenarios currently defined at AN L4. The subscription and publication metamodel (Notification-T) is defined as following.

Subscription Description: It is used by the producer agent (the publisher of network event) to send messages to the consumer agent (the subscriber of a network event), describing the basic requirements for event subscription and notification. Its value is expressed in natural language descriptions. In practical applications, this attribute can directly contain specific or abstract event subscription and notification information. When the subscription and notification information is sufficiently concrete, the producer agent can understand/parse the subscription intent and perform the notification task. When the information is relatively abstract, the consumer agent needs to supplement relevant details in attributes such as Notification Subject, Notification Conditions to facilitate parsing and notification by the producer agent.

- **Attributes:**
 - **Subscription Description:** It is used by the producer agent (the publisher of a network event) to send messages to the consumer agent (the subscriber of a network event), describing the basic requirements for event subscription and notification. Its value is expressed in natural language descriptions. In practical applications, this attribute can directly contain specific or abstract event subscription and notification information. When the subscription and notification information is sufficiently concrete, the producer agent can understand/parse the subscription intent and perform the notification task. When the information is relatively abstract, the consumer agent needs to supplement relevant details in attributes such as Notification Subject, Notification Conditions to facilitate parsing and notification by the producer agent.
 - **Notification Subject** (optional attribute): Used to define the types of events to be subscribed and notification. For event notification types already defined by the current AN L4, the value can be directly filled in for this attribute, such as incident, energy efficiency optimization events, etc. For unknown event, this attribute can be omitted. Its value is described in natural language or structured data.
 - **Notification Conditions** (optional attribute): Defines the conditions for triggering the notification. In practice, this attribute can specify monitored parameters and notification thresholds. Its value is described in natural language or structured data.
 - **Event Data Reporting Format** (optional attribute): Defines the expected format for event reporting to facilitate linking related tasks. Its value is described in natural language or structured data.
 - **Expected Output:** Indicates whether the subscription and notification were successful.

6.4.2.1. Network Event subscription and notification Extension Specification

When using natural language (NL), Notification-T prompt templates are defined based on the Notification-T metamodel for different network events. This section introduces three examples of prompt templates for incident, energy efficiency optimization, and performance optimization. Future versions will feature optimizations and provide additional efficient prompt templates.

An example of a Network Incident subscription and notification Prompt Template is shown below and should be documented in the specification document:

```
## Subscription Description

Please complete the subscription and reporting of autonomous data based on the
following information: event topic, subscription criteria, expected output format, etc.

## Notification Subject

incident

## Notification conditions

Report when the fault level reaches : Primary | Intermediate | critical

## Event Data Reporting Format

Incident reporting should include the following information:

- Basic incident information

Includes message type, fault ID, fault name, fault time, fault type, fault location, etc.

- Incident analysis results

Results of incident analysis and diagnosis, such as root cause of the fault and diagnosis
details.

- Event Impact Level

Includes information such as fault level, impact of the fault, etc.

- Event Handling Recommendations

- Fault Resolution Suggestions

## Expected Output

1. Subscription status, success or failure

2. Reason for subscription failure (optional)
```

An example of an Energy efficiency optimization subscription and notification Prompt Template is shown below and should be documented in the specification document:

```
## Subscription Description

Please complete the subscription and reporting of autonomous data based on the
following Notification Subject, subscription conditions, expected output format, and
other information.

## Notification Subject

energy-efficiency optimization

## Notification Conditions

Daily scheduled push or intent status (satisfied/unsatisfied) change push

## Event Data Reporting Format
```

Energy efficiency optimization event reports should include the following information:

- Basic intent report information
 - Report unique ID, intent reference number
- Detailed intent report information
 - Intent satisfaction information, expected execution result list
- Extended information

Expected Output

1. Subscription status: Success or Failure
2. Reason for subscription failure (Optional)

An example of an Network Quality Optimization Event subscription and notification Prompt Template is shown below and should be documented in the specification document:

Subscription Description

Please complete the subscription and reporting of autonomous data based on the following Notification Subject, subscription conditions, expected output format, and other information.

Notification Subject

network quality optimization

Notification Conditions

Subscribed metrics: Tunnel optimization, Routing policy

Threshold conditions: Base station downlink delay > xx ms, Jitter > xxms

Area scope: Guangdong-Guangzhou

Event Data Reporting Format

1. Reporting content: Network Quality Optimization Report (1. Optimization status, 2. Detailed description of optimization execution results, 3. Comparison of optimized performance metrics, 4. Expansion/planning/renovation suggestions, 5. List of links or objects involved requiring disposal, including object identifiers, reasons for optimization failure and recommended actions, 6. Domain-specific information)

2. File format: CSV format

Expected Output

1. Subscription status: Success or Failure
2. Reason for subscription failure (Optional)

6.4.3. Usage Example: Activating Extensions for Incident Subscription

The A2A extension mechanism permits conveying extension content either in the metadata container or within message parts. For consistency and readability, this document selects the metadata-based form to describe extension activation. Implementations MAY support both mechanisms. Standardized recommendation solutions will be provided in the next version

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/NL/v> Body:

```
"message": {
  "role": "agent",
  "parts": [ {"text": "Subscribing to and reporting of network fault events"}]
"metadata":
{ "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/NL/v1": "## Subscription Description\nPlease complete the subscription and reporting of autonomous data based on the following information: event topic, subscription criteria, expected output format, etc.\n## Notification Subject\nincident\n## subscription conditions\n Report when the fault level reaches critical\n## Notification Data Reporting Format\nThe incident report should include the following information:\n1. Basic Event Information\nIncludes message type, fault ID, fault name, fault time, fault type, fault location, etc.\n2. Incident Analysis Results\nResults of the event analysis and diagnosis, such as root cause of the fault, diagnostic details.\n3. Event Impact Level\nIncludes information such as fault level, impact of the fault, etc.\n4. Event Handling Recommendations\nSuggestions for fault resolution.\n## Expected Output\nsuccess."}
  "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
}
```

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification-T/NL/v1>

The response body is:

```
{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
  },
  "artifacts":
  {
    "artifactId": "artifact-uuid",
```

```

    "parts": [
      {
        "text": "The subscription is successful",
      }
    ]
    "metadata":
    {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Notification
-T/NL/v1": "1. Basic incident Information\nMessage Type: create\nFault ID:
323213\nFault Name: Single-Wave Optical Power Fault\nFault Type: Optical Power
Fault\nFault Location: Network Element: 054c1ba7-e778-4f3a-94cb-f6fcb6ac44f2,
Network Element Type: OptiX OSN 9800 U32\nFault Time: 2025-08-04T16:00:00Z\n2.
Incident Analysis Results\nRoot Cause: Degradation of the internal optical path within
the network element, with a sudden drop of 13.60 dB at the sink port. Source Port:
NE(7-228)-Shelf0-12-12WSMD9-1(IN/OUT), Sink Port: NE(7-228)-Shelf0-5-15WSMD9-
1(IN/OUT).\n3. Impact Level of the Event\nFault Level: Critical\nImpact of the Fault:
Unknown\n4. Event Handling Recommendations\nFault Repair Suggestion: Please
check whether the fiber jumpers, boards, and modules involved in this optical path
within the network element are faulty.}
  }
}
}

```

6.5. Task Negotiation Extension

6.5.1. Overview

The A2A is extended based on Task Negotiation, which encompasses disambiguation of tasks (Target Negotiation), resolution of resource conflicts (feasibility negotiation), and inquiries for necessary information during task execution (information negotiation). Specifically:

Intent-based negotiation aims to address ambiguities in task processing, such as whether to trigger related operations or linked tasks. Conflicts among multiple task objectives will be handled by a Multi-Agent Orchestra [IG1251E].

Feasibility negotiation refers to a discussion about the feasibility of the task objectives, aiming to avoid issues such as resource unavailability—e.g., when the required collaborating agent is currently handling other tasks—or the inability to meet performance requirements, such as when the specified execution time for the task is too short. In such cases, the task initiator proactively initiates negotiation with the task receiver.

During task execution, information inquiries refer to the supplementary requests for essential information required to perform the task. These inquiries are proactively initiated by the task executor toward the task requester. Typically, such information inquiries during task execution serve complaint handling or cross-service (east-west) tasks. The characteristic of these tasks is that the task initiator does not provide some necessary information when issuing the task, requiring the task executor to request supplemental information during the execution process.

The brief overview of this capabilities extension is as follows:

- **Capability Extension:** negotiation-metadata
- A Negotiation-T message **MAY** be expressed in natural language or reference 3GPP specification:
 - a. When using natural language (NL), reference the URI:
<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1>
 - b. When adopting the 3GPP approach, align with **TS 28.312** and use the URI:
<https://projects.tmforum.org/a2aproject/telecommunication/extensions/TASK-T/3GPP-intent/v19.0>

Agent Card JSON Code:

```
{
  "name": "fault-mgmt-agent",
  "description": "Fault Management Agent",
  "url": "http://0.0.0.0:8085",
  "version": "1.0",
  "defaultInputModes": [
    "text"
  ],
  "defaultOutputModes": [
    "text"
  ],
  "capabilities": {
    "streaming": "true",
    "push_notifications": "true",
    "extensions": [
      {
        "uri":
https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-

```

```

T/NL/v1", "https://projects.tmforum.org/a2aproject/telecommunication/extensions/TA
SK-T/3GPP-intent/v19.0",
"description": "Add the intent negotiation extension to the 'metadata' field of the
Message object being sent."
}
]
},
"skills": [
{
"id": "fault",
"name": "Fault Diagnosis",
"description": "Fault detection and root cause diagnosis"
}
],
"supportsAuthenticatedExtendedCard": true
}

```

6.5.2. Negotiation-T Data Model

Based on the three forms of task negotiation, this document defines the metamodel for task negotiation as following.

- Attributes:
 - The following attributes are used for ambiguity resolution of task objectives:
 - **Objective Confirmation:** Used by the task executor agent to send a request to the task requester agent for objective negotiation, addressing ambiguity in the task goal.
 - **Original Objective:** The original goal information sent from the task requester to the task executor agent, with values described in natural language or structured data.
 - **Objective Recognition:** The summarized task description identified by the task executor agent, with values expressed in natural language or structured data.
 - **Follow-Up Information:** When the original goal is ambiguous or lacks necessary details, the task executor agent sends a message to the task requester agent to supplement essential information, with values described in natural language or structured data.

- The following attributes are used for task feasibility negotiation:
 - **Feasibility Confirmation:** Used by the task requester agent to send a request to the task executor agent to assess whether task execution is feasible, requesting the task executor agent to evaluate if the basic requirements can be met. Values are described in natural language or structured data.
 - **Task Information:** Describes the task requirements within the message sent by the task requester agent to the task executor agent, with values in natural language or structured data.
 - **Constraints:** Describes the desired execution requirements of the initial task, including timing, resource availability, etc., with values in natural language or structured data.
 - **Output Format:** Describes the expected output format, including whether the task is feasible, alternative solutions if not feasible, and additional resources needed for task execution or alternatives. Values are described in natural language or structured data.
- The following attributes facilitate dynamic corrections during task execution:
 - **Information Supplement Request:** Used by the task executor agent to request to the task requester agent to provide additional information, sent as a message. The value is described in natural language or structured data. In practical application, this attribute can contain specific or abstract supplementary information needs. When the supplement request is sufficiently concrete and understandable/parsible by the task requester agent, it will evaluate whether to provide the related information. When the request is relatively abstract, the task executor agent needs to include relevant details in attributes such as Required Information Items and Output Format Specification.
 - **Required Information Items (optional):** Describes specific information needed, with values in natural language or structured data.
 - **Output Format Specification (optional):** Describes the expected output format, typically recommending that, if related information can be provided, supplement it; if not, provide alternative requirements. There is no strict constraint. Values are in natural language or structured data.

- **Data Return Confirmation (optional):** Used by the task requester agent to confirm that the related supplementary information can be provided. Values are in natural language or structured data.
- **Data Request Rejection (optional):** Used by the task requester agent to inform the task executor agent that the related supplementary information cannot be provided. Values are in natural language or structured data.
- **Reason for Rejection Explanation (optional):** When the supplementary information cannot be provided, the task requester agent sends a reason to the task executor agent, described in natural language or structured data.
- **Alternative Solution Proposal (optional):** When supplementary information cannot be provided, the task requester agent proposes alternative requirements to the task executor agent, expressed in natural language or structured data.
- **Output Format Specification (optional):** Used by the task requester agent to specify the expected output format, often to indicate whether substitute solutions are accepted, without strict constraints. Values are described in natural language or structured data.

This document defines 3 commonly used task negotiation templates in the ICT field including, Target Negotiation, feasibility negotiation, and information negotiation.

6.5.2.1. Task Negotiation Extension Specification

Network Target Negotiation

When using natural language (NL), an example of a Network Target Negotiation Prompt Template is shown below and should be documented in the specification document:

Intent Confirmation

Please confirm whether the intent recognition result is accurate. If not, provide additional information to clarify the intent.

Original Intent

The original input content (such as questions, commands, requests, etc.)

Intent Recognition Result

A brief summary of the intent

Follow-up information on Intent

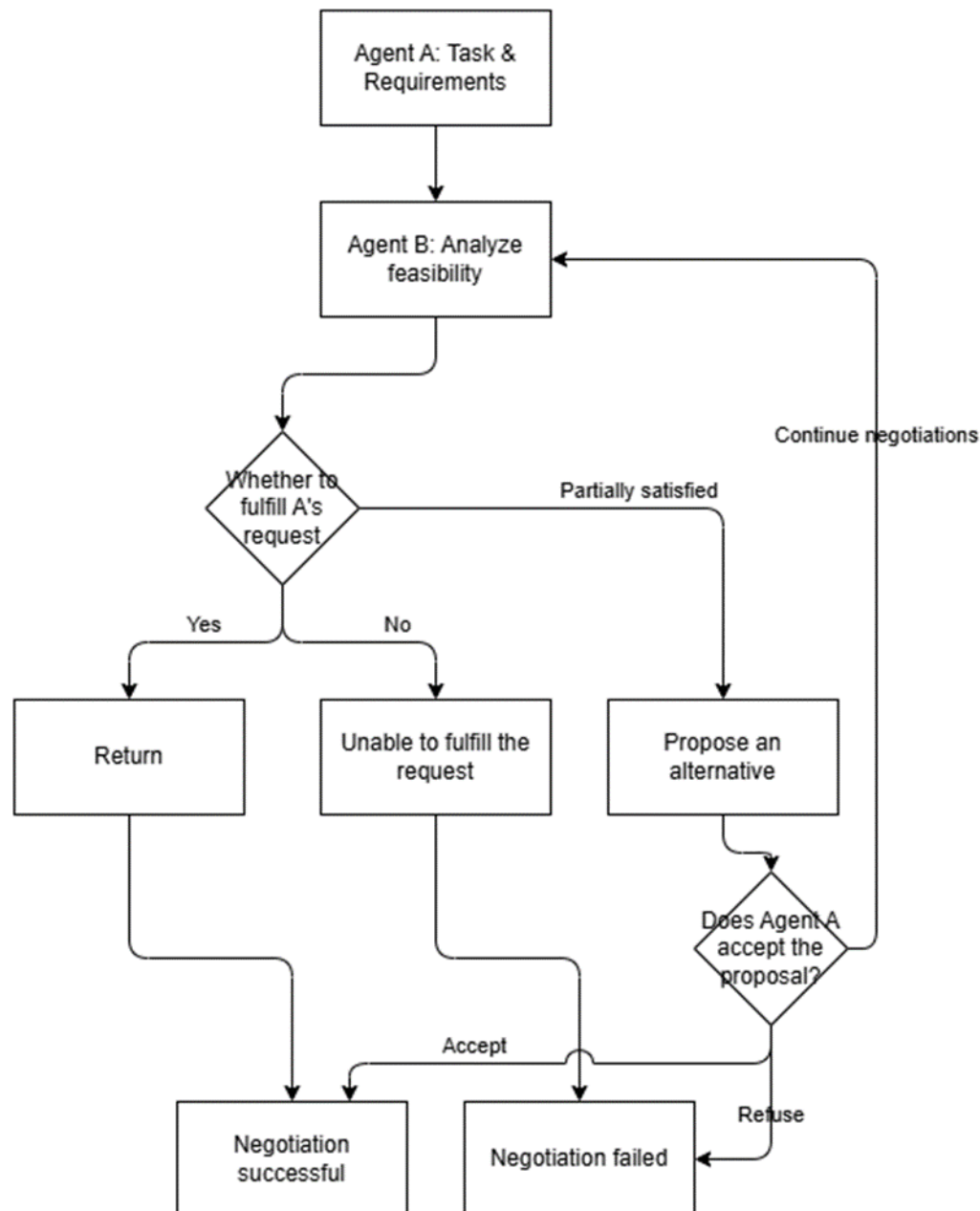
The original intent is ambiguous; please provide precise follow-up information.

Output Format

Please respond with the follow-up information regarding the intent.

Feasibility Negotiation Extension Specification

Feasibility negotiation refers to a discussion about the feasibility of the task objectives, aiming to avoid issues such as resource unavailability or the inability to meet performance requirements. In such cases, the task initiator proactively initiates negotiation with the task receiver, and typically follow the procedure below:



The following is an example of the negotiation process:

Step 1: Agent B receives the task information sent by Agent A, which includes the task objective, xx requirements, etc. Agent B then provides a solution for the task. The solution includes candidate options, each comprising the thinking method, estimated time required for the task, and the essential LLMs or other models needed for the task processing.

Step 2: Agent A receives the solution, including the candidate thinking methods. If the solution is acceptable, Agent A selects one of the recommended thinking methods as the final answer and sends it to Agent B. If it is not acceptable, Agent A updates the task requirement information—such as the required time, essential LLMs, etc.—and sends the updated requirements back to Agent B.

Step 3: Agent B receives the selected reasoning method and executes it to carry out the task.

When using natural language (NL), an example of a Feasibility Negotiation Prompt Template is shown below and should be documented in the specification document:

Feasibility Confirmation

Please evaluate whether the following task requirements are feasible and return the confirmation result; if not feasible, please provide alternative plans or suggestions.

Task Information

Task description, objectives, etc.

Constraints

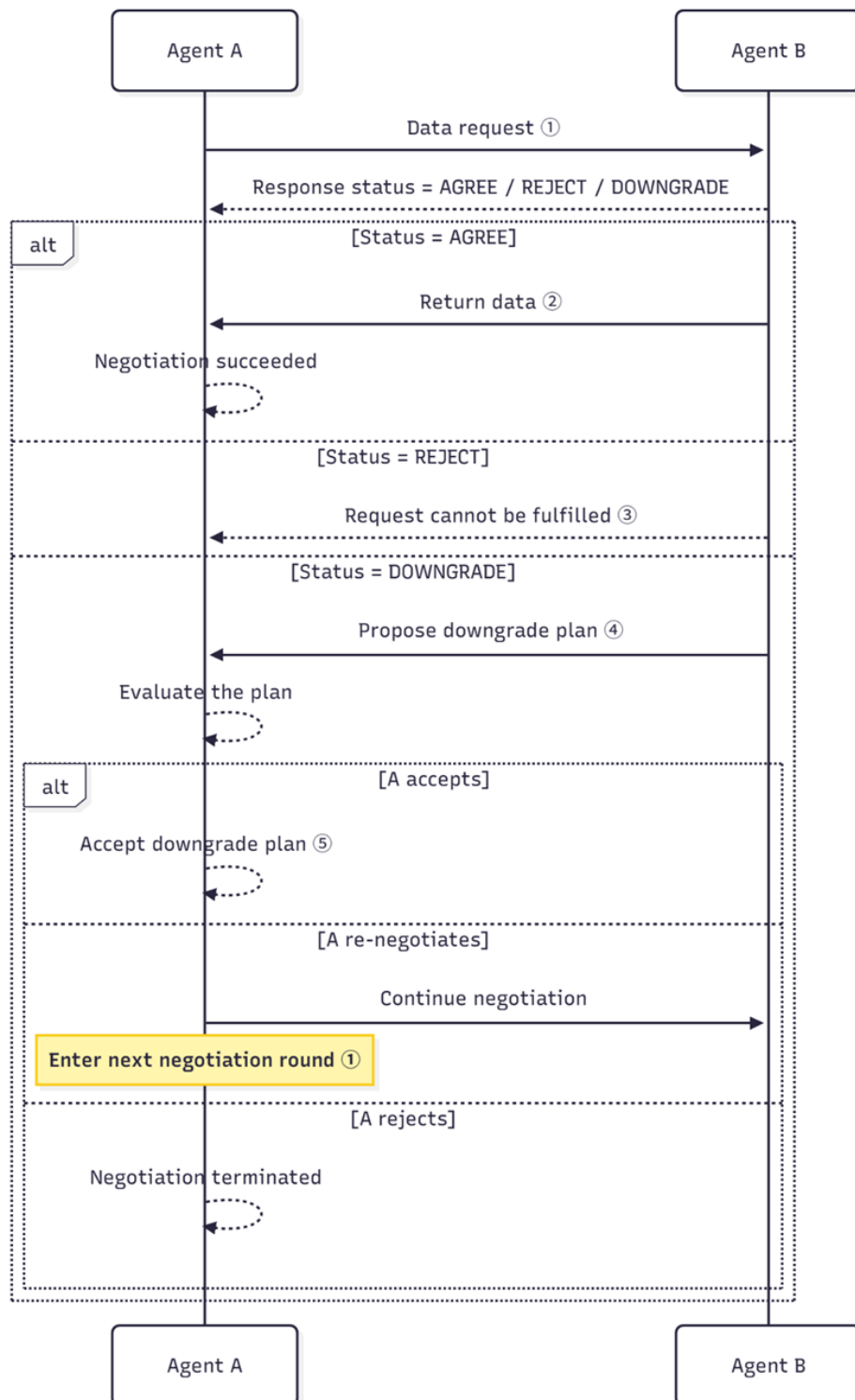
Include execution deadlines, required resources, other constraints, etc.

Output Format

1. Feasibility status: Feasible / Not feasible
2. Explanation for the decision
3. If not feasible: Alternative Suggestions (such as extending the timeline or adjusting resources)
4. Explanation of involved risks or dependencies

Information Negotiation Extension Specification

During the task execution process, information inquiry refers to the supplementary requests for information essential to performing the task. These inquiries are proactively initiated by the task executor to the task sender, and typically follow the procedure below:



When using natural language (NL), an example of an Information Negotiation Prompt Template is shown below and should be documented in the specification document:

Information Supplement Request

The task requires the following information to continue execution. Please provide the relevant content.

Required Information Items

Field, Mandatory or Not, List of Currently Missing Items

Output Format Specification

1. Supplemental field values
2. Remarks
3. If unable to provide, state the reason or alternative solution.

Data Request Confirmation Extension specification Markdown document:

Confirmation

The required information or data files have been provided according to the task requirements. Please confirm receipt.

Data Request Rejection Extension specification Markdown document:

Data Request Rejection

Currently unable to fulfill the requested information requirements. Please review the reason explanation below.

Reason for Rejection Explanation

Includes unavailable data items, primary reasons, scope involved, etc.

Alternative Solution Proposal Extension specification Markdown document:

Alternative Solution Proposal

Currently unable to fully meet the original data requirements. Please indicate whether the following alternative solution is acceptable.

Alternative Solution Content

Includes alternative solution description (field adjustments, scope reduction, precision reduction, etc.)

Output Format Specification

- Feedback (Accept / Re-negotiate / Reject).

6.5.3. Usage Example: Activating Extensions

The A2A extension mechanism permits conveying extension content either in the metadata container or within message.parts. For consistency and readability, this document selects the metadata-based form to describe extension activation. Implementations MAY support both mechanisms. Standardized recommendation solutions will be provided in the next version.

6.5.3.1. Activating Extensions for Target Negotiation

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1>

Body:

```
{
  "message": {
    "role": "agent",
    "parts": [
      {
        "text": "Please confirm whether the intent recognition result is accurate. If not,
provide additional information to clarify the intent. ",
      }
    ]
  },
  "metadata": {
    {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1": "## Intent Confirmation/nPlease confirm whether the intent recognition
result is accurate. If not, kindly provide additional information to clarify the intent./n##
Original Intent/n\"XX community has gone down. Please handle it as soon as possible.\"
/n## Intent Recognition Result/nMain intent: Handle the fault of the XX community
going down./nThe process may include: fault localization (cause analysis), repair
(execute recovery actions), and recovery verification (verify the community's recovery
status)./n## Follow-up information on Intent/nThe current instruction \"Please handle it
as soon as possible\" is ambiguous:/n- Is it only necessary to perform fault localization
and report the cause? /n- Or do you need to perform repair and recovery verification
simultaneously? /n- If repair is required, do you have on-site resource permissions
(such as remote restart or configuration modification)?/n## Output Format Guidelines/
n1. Please respond with the follow-up question regarding the intent.\"}
    "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
  }
}
```

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/V1>

The response body is:

```
{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts": {
      {
        "artifactId": "artifact-uuid",
        "parts": [
          {
            "text": "The information about the supplementary task objective is as follows",
          }
        ]
      }
    },
    "metadata": {
      {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1": "## Follow-up Question on Intent/n- Perform fault localization and report the cause/n- Execute repair actions and recovery verification simultaneously/n- Have on-site resource permissions, including remote restart and configuration modification permissions"}
    }
  }
}
```

6.5.3.2. Activating Extensions for feasibility negotiation

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1>

Body:

```
{
  "message": {
    "role": "agent",
    "parts": [
      {
```

```

      "text": " Please evaluate whether the following task requirements are feasible
and return the confirmation results; if not feasible, please provide alternative plans or
suggestions.",
    }
  ]
  "metadata":
  {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-
T/NL/v1": "## Feasibility Confirmation/nPlease evaluate whether the following task
requirements are feasible and return the confirmation results; if not feasible, please
provide alternative plans or suggestions./n## Task Information/nTask Description:
Automate diagnosis of "Xi'an area XX cluster base station failure group," identifying the
affected scope and root cause types./nTask Objectives: Based on current alarms and
performance metrics, quickly identify the cause of the group failure (such as
transmission interruption, power failure, or software anomalies) and output a
preliminary diagnosis report./n## Constraints/n- Execution deadline: complete the
preliminary diagnosis within 10 minutes./n- Required resources: real-time alarm
streams, transmission topology information./n## Output Format/n1. Feasibility status:
Feasible / Not feasible/n2. Explanation for the decision: (e.g., currently executing other
tasks, unable to meet the time constraints, resources insufficient)/n3. If not feasible:
Alternative Suggestions (such as relaxing time requirements, adjusting resources)/n4.
Explanation of involved risks or dependencies"}
    "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
  }
}

```

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T>

The response body is:

```

{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts":
    {
      "artifactId": "artifact-uuid",

```

```

    "parts": [
      {
        "text": "The task cannot be performed. The related information and
alternative solutions are as follows",
      }
    ]
    "metadata":
{"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-
T/NL/v1": "1. Feasibility Status: Not feasible/n2. Explanation for the decision: The Agent
Bar required for cooperation is currently executing other tasks./n3. Alternative
Suggestions: Have Agent Foo substitute for Agent Bar to execute the task, or initiate the
task again after 10 minutes."}
  }
}
}

```

6.5.3.3. Activating Extensions for Information negotiation

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/v1>

Body:

```

{
  "messageId": "b3a1f2e4-6c9a-4d87-9f1a-3e7c9a5f8b21",
  "message": {
    "role": "agent",
    "parts": [
      {
        "text": "Accurate network target values are required for task execution.",
      }
    ]
  }
  "metadata":
{"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-
T/NL/v1": "## Information Supplement Request\nThe task requires the following data to
locate the vehicle cross-domain communication interruption issue.\n\n## Required
Information Items\n| Field | Mandatory | Currently Missing |\n|:---|:---|:---|\n| Cross-
domain switch log | Yes | None |\n| Core network parameter table | Yes | None |\n|
Wireless measurement data | No | Only within this domain |\n\n## Output Format

```

Specification\n1. Supplemental field values\n2. Remarks\n3. If unable to provide, state the reason or alternative solution."}

}

}

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1>

The response body is:

```
{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts": [
      {
        "artifactId": "artifact-uuid",
        "parts": [
          {
            "text": "Unable to fully meet the original data requirements. Please refer to the following options.",
          }
        ]
      }
    ],
    "metadata": {
      {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1": "## Alternative Solution Proposal\nCurrently unable to fully meet the original data requirements. Please refer to the following options.\n## Alternative Solution Content\n| Original Field | Available Solution | Alternative Explanation |\n|:----|:----|:----|\n| Cross-domain switch log | Provide summarized statistics (success rate, failure reasons) | De-identification restrictions |\n| Core network parameter table | Only provide mappings between this domain and boundary AMF | Reduced accuracy |\n| Wireless measurement data | 5-minute granularity average values | Low-precision samples |\n\n## Output Format: Accept / Re-negotiate / Reject"}
    }
  ]
}
```

Or provide structured data like:

```
{
  "task": {
    "id": "f7c8d9e0-1234-4abc-8d12-abcdef987654",
    "status": {
      "state": "completed"
    },
  },
  "artifacts": [
    {
      "artifactId": "a1b2c3d4-5678-4e9f-8abc-123456abcdef",
      "parts": [
        {
          "text": "Alternative Solution Content: the network targets for the music festival assurance are as follows",
        },
        {
          "data": {
            "Intent": {
              "Id": "intent_8",
              "userLabel": "Radio_Service_Deliver",
              "intentExpectation": [
                {
                  "expectationTargets": [
                    {
                      "targetName": "ULLatency",
                      "targetCondition": "IS_LESS_THAN",
                      "targetValueRange": "15"
                    },
                    {
                      "targetName": "ULThptPerUE",
                      "targetCondition": "IS_GREATER_THAN",
                      "targetValueRange": "100"
                    },
                    {
                      "targetName": "DLThptPerUE",
                      "targetCondition": "IS_GREATER_THAN",

```

```

        "targetValueRange": "300"
      }
    ]
  }
]
}
}
}
]
}
]
}
}

```

Task executor feedback (accepting alternative solutions)

Body:

```

{
  "message": {
    "role": "agent",
    "parts": [
      {
        "text": "Accept the alternative solution"
      }
    ]
  },
  "metadata": {
    "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation-T/NL/v1": "Accept."
  },
  "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
}

```

Task publisher returns a negotiation success message:

```

{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {

```

```

    "state": "completed"
  },
  "artifacts": [
    {
      "artifactId": "artifact-uuid",
      "parts": [
        {
          "text": " The required information or data files have been provided according
to the task requirements. Please confirm receipt.",
        }
      ]
    }
  ],
  "metadata": {
    "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Negotiation
-T/NL/v1": "## Confirmation\nThe required information or data files have been provided
according to the task requirements. Please confirm receipt.\n\n| File | Description |
Transfer Method |\n|:---|:---|:---|\n| HO_Summary.csv | Cross-Domain Switch
Statistics | shared_storage |\n| AMF_Mapping.json | Core Network Mapping Table |
api_transfer |\n\nNegotiation Result: AGREE."
  }
}

```

6.6. Security Extension

6.6.1. Overview

In the telecom domain, any high-risk operation (e.g., configuration changes) requires manual or high-privilege system approval, with full audit trails retained. Such operations are governed by a standardized approval and authorization process. Moreover, the business boundaries of agents across different layers, domains, and functions must be explicitly defined, and CSPs shall be permitted to customize or configure these boundaries.

This document aims to define a data model for O&M tasks in the telecommunications to support secondary validation of AI-generated knowledge, and to address dynamic requests for operational permissions. It focuses specifically on content-related security threats from the OWASP framework, namely T1, T5, T6, and T15. Specifically, this includes:

- A **Prompt template for configuration operation permission policy**, used by CSPs to customize the business boundaries of agents;

- A **Prompt template for modifying configuration operation policy**, used by CSPs to modify the business boundaries of agents;
- A **Prompt template for Dynamic Authorization Request during execution**, used by agents to dynamically request operation permissions from high-privilege personnel or systems.

In practice, it is recommended to define the default authorization policy for unconfigured operations as **"request on each use"**.

For issues related to agent authentication and the hardening of secure connections, please refer to GB1087.

The brief overview of this capabilities extension is as follows:

- **Capability Extension:** Authorization-metadata
- An Authorization-T message **MAY** be expressed in natural language, reference the URI:
<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

Agent Card JSON Code:

```
{
  "name": "fault-mgmt-agent",
  "description": "Fault Management Agent",
  "url": "http://0.0.0.0:8085",
  "version": "1.0",
  "defaultInputModes": [
    "text"
  ],
  "defaultOutputModes": [
    "text"
  ],
  "capabilities": {
    "streaming": "true",
    "push_notifications": "true",
    "extensions": [
      {
        "uri": "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1",
        "description": "Add the dynamic Authorization extension to the 'metadata' field of the Message object being sent."
      }
    ]
  }
}
```

```
]
},
"skills": [
{
  "id": "fault",
  "name": "Fault Diagnosis",
  "description": "Fault detection and root cause diagnosis"
}
],
"supportsAuthenticatedExtendedCard": true
}
```

6.6.2. Authorization-T Data Model

Based on the three forms of task Authorization, this document defines the metamodel as following.

- Attributes:
 - The following attributes are used for the configuration or modification of operation permission policies:
 - **Task Description:** This attribute is used to describe the necessary information required for an agent to configure or modify an authorization policy. Typically, an upper-layer agent or personnel with configuration privileges sets the operational permissions for lower-layer agents, thereby defining clear business boundaries. The value of this parameter is presented in the form of a natural language description.
 - **Operation List:** This parameter is used to describe the operational permission information that a lower-layer agent is allowed to handle during the execution of operations and maintenance management tasks. Specifically, it includes the operation name, operation type, authorization policy, validity period, allowed time window, trigger execution conditions, etc.
 - **Expected Output:** Indicates whether the configuration or modification succeeded. If it fails, the reason for the failure is returned.

- The following attributes are used for the Dynamic Authorization Request during execution:
 - **Task Description:** This attribute specifies the essential information needed in the process of dynamically requesting permissions, thus solving the approval challenge associated with high-risk operations. The value of this parameter is presented in the form of a natural language description.
 - **Operation permission application list.:** This attribute enables a lower-layer agent to dynamically request, from a higher-privilege agent or a human, both the permissions and the necessary task-related information required to complete a task during execution. This includes, but is not limited to, the operation name, operation type, operation object, execution reason and context, and a summary of the implementation plan.
 - **Expected Output:** Indicates whether the dynamic authorization request succeeded. If it fails, the reason for the failure is returned.

This document defines 3 commonly used task negotiation templates in the ICT field including, configuration operation permission policy, modifying configuration operation policy, Dynamic Authorization Request during execution.

6.6.2.1. Authorization-T Extension Specification

Configuration Operation Permission Policy

An example of a Configuration Operation Permission Policy Prompt Template is shown below and should be documented in the specification document:

Task Description

Please configure a complete authorization policy and execution constraints for each operation based on the <Operation List>, and return the task response result according to the structure defined in the <Expected Output>. The <Operation List> represents a list of operation permission objects, including elements such as operation name, operation type, authorization policy, validity period, allowed time windows, and trigger conditions. The <Expected Output> indicates the expected content to be returned.

Operation List

{{Operation Permission Object}}

Expected Output

Return whether the policy configuration was successful. If unsuccessful, provide the reason for the failure.

Configuration Modify Operation Policy

An example of a Configuration Modify Operation Policy Prompt Template is shown below and should be documented in the specification document:

Task Description

Please modify the authorization rule configuration for the specified operations based on the <Operation List>. The modification scope includes the operation's authorization policy, validity period, execution constraints, and other related elements. Return the task response result according to the structure defined in the <Expected Output>. The <Operation List> represents a list of operation permission objects, including elements such as operation name, operation type, operation object, authorization policy, validity period, allowed time windows, and trigger execution conditions. The <Expected Output> indicates the expected content to be returned.

Operation List

{{Operation Permission Object}}

Expected Output

Return whether the modification was successful. If unsuccessful, provide the reason for the failure.

Dynamic Authorization Apply

An example of a Dynamic Authorization Apply Prompt Template is shown below and should be documented in the specification document:

Task Description

Please provide the approval result for the specified operation authorization request based on the < Operation permission application list>, and return the task response result according to the structure defined in the <Expected Output>. The < Operation permission application list> includes elements such as operation name, operation type, operation object, execution reason and context, change ticket number, and a summary of the implementation plan document. The <Expected Output> indicates the expected content to be returned.

Operation permission application list

{{Operation Permission Object}}

Expected Output

The approval result should include the following information:

1. Result: Allow or Deny
2. Reason for denial (optional)
3. Validity period: Authorization expiration time in UTC
4. Authorization credential
5. Approver

6.6.3. Usage Example: Activating Extensions

This chapter introduces how to activate and utilize the Authorization extension for telecommunications domain.

- **Activating Extensions for Configuration Operation Permission Policy**
-

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

Body:

```
{
  "message": {
    "role": "agent",
    "parts": [ { "text": "" } ]
    "metadata": {
      {"https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1": "## Task Description/nPlease configure a complete authorization policy and execution constraints for each operation based on the <Operation List>, and return the task response result according to the structure defined in the <Expected Output>. The <Operation List> represents a list of operation permission objects, including elements such as operation name, operation type, operation object, authorization policy, validity period, allowed time windows, and trigger execution conditions. The <Expected Output> indicates the expected content to be returned./n/n## Operation List/nOperation 1:/nOperation Name: Open peer port/nOperation Type: Port software category/nOperation Object: NE 3600HWCSPPRT1c2318e9c54e3d2a/nAuthorization Policy: Apply each time/nTrigger Execution Condition: Triggered when a fault is detected/n/n## Expected Output/nReturn whether the setup was successful. If unsuccessful, provide the reason for the failure./n"},
      "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
    }
  }
}
```

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

The response body is:

```
{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts":
    { "artifactId": "artifact-uuid",
      "parts": [ { "text": "The Result of Configuration Operation Permission Policy task
are shown as follows" } ],
      "metadata":
{ "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1": " success" }
    }
  }
}
```

- **Activating Extensions for Configuration Modify Operation Policy**

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

Body:

```
{
  "message": {
    "role": "agent",
    "parts": [ { "text": "" } ]
    "metadata":
{ "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1": "## Task Description/n Please modify the authorization rule
configuration for the specified operations based on the <Operation List>. The
modification scope includes the operation's authorization policy, validity period,
execution constraints, and other related elements. Return the task response result
according to the structure defined in the <Expected Output>. The <Operation List>
represents a list of operation permission objects, including elements such as operation
name, operation type, operation object, authorization policy, validity period, allowed
time windows, and trigger execution conditions. The <Expected Output> indicates the
expected content to be returned./n/n## Operation List/nOperation 1:/nOperation
Name: Open peer port/nOperation Type: Port software category/nOperation Object: NE
3600HWCSOCPRT1c2318e9c54e3d2a/nAuthorization Policy: Authorized/nValidity
Period: 2026-10-05T23:59:59Z/nAllowed Execution Time Window: 00:00-02:00/nTrigger
```

Execution Condition: Triggered when a fault is detected/n/n## Expected Output/nReturn whether the modification was successful. If unsuccessful, provide the reason for the failure."},

```
"messageId": "9229e770-767c-417b-a0b0-f0741243c589"
}
}
```

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

The response body is:

```
{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts":
    { "artifactId": "artifact-uuid",
      "parts": [ { "text": "The Result of Configuration Modify Operation Policy task are shown as follows", } ]
    },
    "metadata":
    { "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1": "success" }
  }
}
```

- **Activating Extensions for Dynamic Authorization Apply**

An additional field X-A2A-Extensions needs to be added to the request header, with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

Body:

```
{
  "message": {
    "role": "agent",
    "parts": [ { "text": "Please approve the following operation permission request", } ]
    "metadata": {
      "https://projects.tmforum.org/a2aproject/telecommunication/extensions/
      Authorization-T/NL/v1": "## Basic Operation Information/nOperation Name: Remotely
      open peer port/nOperation Type: Port software category/nOperation Object: NE
      3600HWCSOCPRT1c2318e9c54e3d2a/nExecution Reason and Context: xxx/nChange
      Ticket Number: xxx/nImplementation Plan Document Summary: xxx/n## Expected
      Output/nThe approval result should include the following information:/n1. Result: Allow
      or Deny/n2. Reason for denial (optional)/n3. Validity period: Authorization expiration
      time in UTC/n4. Authorization credential/n5. Approver"},
      "messageId": "9229e770-767c-417b-a0b0-f0741243c589"
    }
  }
}
```

The header of the response body contains the X-A2A-Extensions field with the value:

<https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1>

The response body is:

```
{
  "task": {
    "id": "task-uuid",
    "contextId": "context-uuid",
    "status": {
      "state": "completed"
    },
    "artifacts": {
      "artifactId": "artifact-uuid",
      "parts": [ { "text": "The Result of application are shown as follows", } ]
      "metadata": {
        "https://projects.tmforum.org/a2aproject/telecommunication/extensions/Authorization-T/NL/v1": "1. Result: allow. 2. Validity period: 2025-08-20T11:00:00Z. 3. Authorization credential: xxx. 4. Approver: Michael"
      }
    }
  }
}
```

7. Terms & Abbreviations Used within this Document

7.1. Terminology

Term	Definition	Source

7.2. Abbreviations & Acronyms

Abbreviation/Acronym	Abbreviation/Acronym Spelled Out	Definition	Source

8. References

Reference	Description	Source	Brief Use Summary

9. Administrative Appendix

9.1. Document History

9.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	21-Nov-2025	Alan Pope	Final edits prior to publication
2.0.0	20-Jan-2026	Alan Pope	Final edits prior to publication
3.0.0	29-May-2026	Benson Yap	Final edits prior to publication

9.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	21-Nov-2025	Alan Pope	Initial release
Pre-production	29-Dec-2025	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	20-Jan-2026	Alan Pope	Updated to v2.0.0
Pre-production	02-Mar-2026	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	29-May-2026	Benson Yap	Updated to v2.1.0

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